

Sankalp Singh Bais

+917647834696 • bash.sankalp@gmail.com • linkedin.com/in/bash-sankalp • https://github.com/bash-sankalp

DevOps Engineer | Cloud Infrastructure, CI/CD & Edge Systems

TECHNICAL SKILLS

CI/CD & Cloud Infrastructure: Docker, GitLab CI/CD, Ansible, Linux, Bash

Cloud & Networking: Microsoft Azure (Container Apps, Container Registry, App Services, PostgreSQL, Blob Storage, Entra ID, IAM, RBAC, VNet), Dokploy, WireGuard, ZeroTier

Monitoring & Observability: Prometheus, Grafana, Loki, Node Exporter

Programming & Systems: Python, Flask, Celery, Redis, REST APIs, FFmpeg

Edge & Inference: Raspberry Pi 4/5, Jetson Nano, Hailo SDK, ONNX Runtime, OpenCV

EXPERIENCE

Software Development Engineer

August 2025 - February 2026

Containova Systems LLP

Navi Mumbai, India

- Led **DevOps architecture and operations** for a production edge-based Yard Management System deployed across **40+ industrial devices**
- Designed and maintained **5+ GitLab CI/CD pipelines** to build, containerize, and deploy services to **Azure Container Apps** and **on-prem** infrastructure
- Migrated from Azure App Services to **Azure Container Apps**, reducing cloud costs by **60–70%** and improving rollback reliability
- Transitioned to a **hybrid infrastructure** (on-prem via Dokploy + Azure for PostgreSQL and Blob Storage), reducing vendor lock-in and recurring costs
- Implemented secure **OTA update mechanism** integrated with CI/CD, eliminating manual fleet deployments
- Built centralized **monitoring and logging stack (Prometheus, Grafana, Loki, Node Exporter)** for fleet-wide visibility and faster **incident resolution**
- Designed and migrated to self-hosted **WireGuard VPN architecture**, improving network stability and reducing cloud overhead
- Automated device provisioning via hardened **Bash scripts**, reducing configuration time from **3 hours to 30 minutes**
- Integrated production alerting using **Zoho Cliq API** for real-time failure notifications

Trainee Engineer - Hardware, Software & DevOps

June 2024 - July 2025

Containova Systems LLP

Navi Mumbai, India

- **Containerized** ML training and edge deployment workflows using **Docker**, enabling reproducible builds
- **Re-architected** backend from threaded Flask to **Celery + Redis**, enabling reliable multi-device concurrency
- Migrated edge runtime from Jetson Nano to Raspberry Pi 5 with hardware acceleration, improving inference latency from **~1s/frame** to **~0.03s/frame**
- Optimized edge inference and OCR pipeline, reducing processing latency by over **10x**
- Engineered **real-time RTSP ingestion module** to guarantee latest-frame processing for inference pipeline
- **Self-hosted** and administered **GitLab**, including repositories, runners, and CI configuration
- Designed secure device-to-backend communication over WiFi/Cellular using **VPN-based networking**

Hardware and Software Intern

July 2023 - June 2024

Containova Systems LLP

Navi Mumbai, India

- Designed and built initial edge prototype for an industrial **Yard Management System** providing precise container location tracking
- Architected dual-unit edge system integrating compute, networking, GPS, and PoE camera stack
- Developed **Python-based edge application** for real-time RTSP ingestion and inference
- Implemented **high-precision GPS** ingestion service exposing parsed coordinates via HTTP APIs
- Assisted in hardware selection and development of production-ready deployable units

PROJECTS

System Monitoring Dashboard (Personal Initiative)

August 2025

- Designed and built a **centralized monitoring stack** using **Prometheus, Grafana, and Loki** as a personal initiative, later **deployed across 40+ production edge devices**

- Used to **monitor uptime, resource utilization**, and **stream application logs** securely across multiple client environments over an isolated **VPN**

EDUCATION

B.Tech. Computer Science and Engineering

Shri Shankaracharya Technical Campus • GPA: 8.62

Bhilai, Durg • November 2020 - May 2024